TRANSPORT AND TELECOMMUNICATION INSTITUTE

# Software Engineering

Course Project
Topic: Software development(your topic)

Student name:
Student surname:
St. code:

**RIGA**

# THE TASK FOR COURSE PROJECT

**STEP 1:** Describe the requirements (specification) for software project (what the system should do and how, all functions of a supposed future program). It is assumed that for the realisation of, you should use an object-oriented language.

You should represent:

- **Functional requirements**
- **Minimal system requirements**
- **System users**

**Specification Example:**

> ***University Specification***
>
> *A lecturer, identified by his or her number, name and room number, is responsible for organising a number of course modules. Each module has a unique code and also a name and each module can involve a number of lecturers who deliver part of it. A module is composed of a series of lectures and because of economic constraints and common sense, sometimes lectures on a given topic can be part of more than one module. A lecture has a time, room and date and is delivered by a lecturer and a lecturer may deliver more than one lecture. Students, identified by number and name, can attend lectures and a student must be registered for a number of modules. We also store the date on which the student first registered for that module. Finally, a lecturer acts as a tutor for a number of students and each student has only one tutor.*

**See some examples if you need:**

http://www.uml-diagrams.org/bank-atm-uml-use-case-diagram-example.html?context=uc-examples

Develop a visual system model in terms of rational unified process:

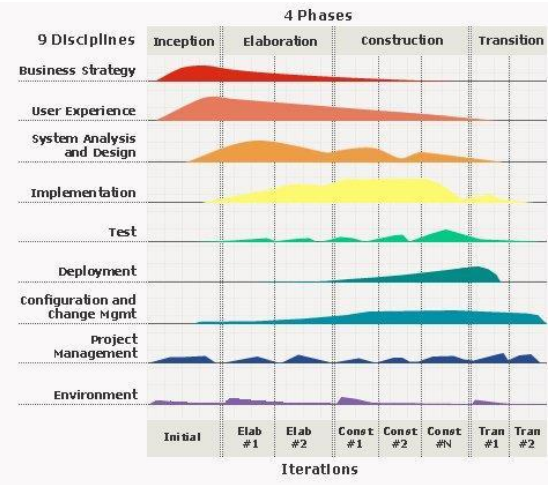


Fig.1 Rational Unified Process

http://theexplorationofmyworlds.wordpress.com/2012/07/02/rup-rational-unified-process/

http://alvinalexander.com/java/java_oo/java_oop_tutorial_1_6Rational_Unified_Process.shtml

**Note:** For model development use any case system (for example in class room Run the CASE system Rational Rose Enterprise Edition press Start / All Programs / Programming / IBM Rational / Rational Rose Enterprise Edition, then press cancel button. For visual model use UML language http://msdn.microsoft.com/en-us/library/dd409432.aspx

**STEP 2**

**INCEPTION PHASE** /see represented fig.1/

- Create and represent Business Use Case diagram (20%) in report (from business view http://www.uml-diagrams.org/use-case-diagrams.html). Describe in report all its elements (specification of actors and use case elements).

**STEP 3**

**ELABORATION PHASE** /see represented fig.1/

- Create and represent System Use Case diagram in report (in terms of functionality view) http://www.uml-diagrams.org/use-case-diagrams.html). Describe in report all its elements-specification for actors and use case elements. See the given examples below:

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**Actor Specification Card**

| *Actor Specification* | | |
|---|---|---|
| **Actor Name:** Customer | | |
| **Type:** Primary | **Personality:** Initiator | **Abstract:** No |
| **Role Description:** A customer is a person who has opened an account with the bank. The customer is the main reason for the existence of ATM machines. The customer interacts with the ATM to obtain convenient banking services at times when he/she cannot or is not convenient to obtain such services at the bank's premises. | | |
| **Actor Goals** (no need to fill this box for secondary actors): • Withdraw cash • Deposit funds • Make transfers • Inquire balances | | |
| **Use Cases Involved with** (events that fulfill goals above; a goal will map to one or more use cases; a use case will map to one or more primary actor goals; no need to fill this box initially, only after use cases have been indentified): • Withdraw funds • Deposit funds • Manage account (make transfers, inquire balances) | | |

21

A **use case specification** represents the sequence of events along with other information that relates to this use case. A typical use case specification template includes the following information:

- Description
- Pre- and Post- interaction condition
- Basic interaction path
- Alternative path
- Exception path

## Example:

| **Overview** | |
|---|---|
| **Title** | [Title of the basic flow use case] |
| **Description** | [Short description of the basic flow] |
| **Actors and Interfaces** | [Identifies the Actors and Interfaces to components and services that participate in the use case] |
| **Initial Status and Preconditions** | [A pre-condition (of a use case) is the state of the system that must be present prior to a use case being performed] |
| **Basic Flow** | |
| STEP 1: …<br>STEP 2: … | |
| **Post Condition** | |
| [A post-condition (of a use case) is a list of possible states the system can be in immediately after a use case has finished] | |
| **Alternative Flow(s)** | |
| [Alternative flows are described here if needed] | |

**Table 2 An example of a use case.**

| **Name** | Handle Replies | |
|---|---|---|
| **Realized User Task** | Manage Interaction Among Participants | |
| **Initiating Actor** | Meeting Facilitator | |
| **Participating Actors** | Meeting Participant | |
| | | |
| **Flow of events** | **Actors** | **System** |
| | 1. The Meeting Facilitator selects "Handle Replies" for a meeting and a question. | |
| | | 2. The system checks if all participants replied [Exception: Slow participant]. |
| | | 3. The system starts the "Close Question Service" and notifies the Meeting Initiator accordingly. |
| **Exceptions** | [Slow participant] The meeting facilitator decides whether to remind the participants or to close the question. In the first case s/he selects the "Remind Participant Service". In the second case s/he selects the "Close Question Service". | |
| **Precondition** | The meeting Initiator has initiated the meeting and asked some question. | |
| **Postcondition** | The participants have been reminded or the question is closed. | |
| **Includes Use Cases** | - | |
| **Used Services** | Check Participant Replies, Remind Participant, Close Question | |
| **Non-functional Requirements** | Response Time, Minimize Amount of Messages, Flexibility | |

## Business Use Case Description

| | | | |
|---|---|---|---|
| **Use Case ID:** | | | |
| **Use Case Name:** | | | |
| **Process Owner:** | | **Last Updated By:** | |
| **Date Created:** | | **Date Last Updated:** | |
| **Business Actor:** | | | |
| **Description:** | | | |
| **Preconditions:** | | | |
| **Postconditions:** | | | |
| **Performance Goal:** | | | |
| **Basic Workflow:** | | | |
| **Alternative Workflow:** | | | |
| **Category:** | | | |
| **Risks:** | | | |
| **Possibilities:** | | | |
| **Special Requirements:** | | | |
| **Assumptions:** | | | |
| **Notes and Issues:** | | | |

- Create and represent sequence diagram for each use case element (at least one for each use case element). Show and comment all sequence diagrams in report, also represent all their scenarios. Represent initial Class Diagram. Initialquality you should represent in the table below:

| **Metric** | **Class name** | **Class name** | **Class name** | **Class name** | **Class name** | **Class name** | **Class name** | **Average value** |
|---|---|---|---|---|---|---|---|---|
| WMC | **value** | … | … | … | … | … | … | … |
| NOC | **value** | … | … | … | … | … | … | …. |
| **System metrics** | | | | | | | | |
| DIT | | | | | | | | |
| NC | | | | | | | | |
| NOM | | | | | | | | |

- Describe and represent in report the plan of next construction phase. Plan 3 iterations of project for supposed realisation. It means that it's necessary to describe which functional possibilities of project you need to realize
- **primarily,**

- **secondary and**
- **thirdly** (describe their priority).

**STEP 4**

**CONSTRUCTION PHASE** /see represented fig.1/

- Create and represent Class diagram on each iteration phase. Comment in report all its elements (classes and relationships between them).
- Represent in report software quality (of class diagram) on each iteration. For representing use class-oriented metrics "The Chidamber and Kemerer Suite" (WMC, DIT, NOM, CBO, RFC, LCOM), see the given table below:

| **Metric** | **Class name** | **Class name** | **Class name** | **Class name** | **Class name** | **Class name** | **Class name** | **Average value** |
|---|---|---|---|---|---|---|---|---|
| **WMC** | value | … | … | … | … | … | … | … |
| **NOC** | value | … | … | … | … | … | … | …. |
| **CBO** | value | | | | | | | |
| **RFC** | value | | | | | | | |
| **LCOM** | value | | | | | | | |
| **SYSTEM METRICS** | | | | | | | | |
| **DIT** | | | | | | | | |
| **NC** | | | | | | | | |
| **NOM** | | | | | | | | |
| **LOC** $\Sigma$ | | | | | | | | |

- Represent in report **Component Diagram** and describe all its elements (which classes were assigned to which components).

**STEP 5:** Make conclusion about course project and about the received quality assessment (is it good or bad). Files of report and visual model (project files) upload to the Moodle system.

Class metrics averages at different iterations

| **Metric** | **Iteration 1** **Average values** | **Iteration 2** **Average values** | **Iteration 3** **Average values** |
|---|---|---|---|
| WMC | | | |
| NOC | | | |
| CBO | | | |
| RFC | | | |
| LCOM | | | |

| System metrics | | | |
|---|---|---|---|
| DIT | | | |
| NC | | | |
| NOM | | | |
| LOC Σ | | | |

**STEP 6**

Represent the screenshots of your program.

Make conclusion…

Finish your report and upload it to the Moodle system.